

Faculty of Engineering

Topics in Computer Science Multi-Agent System

PhD. Willy Ugarte Rojas
willyugarte@gmail.com



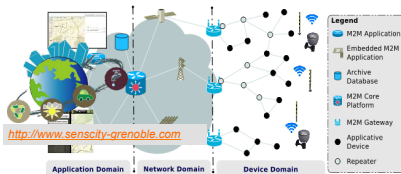
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Open & Decentralized AI Systems

Open & Decentralized AI Systems

Agile governance of M2M infrastructure [Persson et al., 2012]



Supporting human organizational awareness [de Brito et al., 2016]



Social Web of things [Ciortea et al., 2017]



- Strong inter-connection of physical, digital and social worlds to build long-lived socio-technical and cognitive systems
- Openness, distribution, no centralization, intensive use of knowledge

Requirements

Open & Decentralized AI Systems

- ▶ Openness, non centralization & distribution
- ▶ Situated into dynamic and complex environments
- ▶ Representation and reasoning on various types of knowledge
 - ▶ e.g. expert knowledge, policy/norms, legal knowledge, ...
- ▶ Flexible micro-macro loops combining local and global computations
- ▶ Modularity, extensibility, reusability, ...

How to **engineer** such applications?

Definitions

An organisation of autonomous agents interacting with each other within a shared environment

- ▶ **Agents:** autonomous entities able to react to events while pro-actively defining goals and directing actions to achieve them
~ (soft/hard)ware, (coarse/small)-grain, (hetero/homo)geneous,
- ▶ **Environment:** shared medium providing the surrounding conditions for agents to exist and act (e.g. comm. and coord. infra., topology of spatial domain, support of an action model)
~ virtual/physical, passive/active, deterministic or not, ...
- ▶ **Interaction:** motor of dynamic and interoperability in the MAS
~ direct communicative / indirect actions through the environment
- ▶ **Organisation:** abstractions to declare and make accessible to agents their expected collective structure and functioning in a shared environment (e.g. coordination and regulation activities)
~ pre-defined/emergent, static/adaptive, open/closed, ...

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An organisation of autonomous agents interacting with each other within a shared environment

MAS **is more than** a simple set of agents

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Principles

Definitions

Multi-agent system principles

- ▶ **Distribution** of knowledge, resources, reasoning/decision capabilities
- ▶ **Decentralisation** of control, authority (loosely coupled agents)
- ▶ **Coordination** & **Regulation** models and mechanisms to install coordination & regulate the autonomous agents
- ▶ Flexible interlacement of **emergent**, **social order**, **normative** functioning

Agent principles

- ▶ **Situated**, **Reactive**, **Pro-Active**, **Social** & **Organization-aware** entities
- ▶ **Autonomy**: agents may exhibit activities that are not the one expected by the other agents in the system
- ▶ **Delegation**: agents may receive some control over their activities

Some History

Positioning

- ▶ 1973: Distributed Artificial Intelligence (DAI)
 - ▶ Hearsay II (1973): blackboard architecture for speech recognition
 - ▶ Actor Languages (1973): messages as control structures
 - ▶ Beings (1975), Society of Minds (1978)
- ▶ 1980: Agents in the Artificial Intelligence (AI) area
 - ▶ From AI to DAI $\leadsto$ to Decentralized AI (DzAI) $\leadsto$ to Multi-Agent Systems (MAS)
- ▶ 1990: Agents invade other domains
 - ▶ Personal Assistants, Avatars, Mobile Agents, Reactive Agents, ...
- ▶ 1995: Agents spread in other domains, Application domains are enlarging
 - ▶ Artificial Life, Economic Agents, ...
 - ▶ **1st ICMAS Conference in 1996**
- ▶ 2000 -:
 - ▶ Web, Ambient Computing, Internet of Things, ...
 - ▶ **Since 2002, AAMAS Conferences**

Inter-disciplinary Domain

Positioning

- ▶ Direct Links with:
 - ▶ Programming, Object Oriented Programming ...
 - ▶ Artificial Intelligence
 - ▶ Distributed Systems, Parallelism, Concurrent Programming
- ▶ But also:
 - ▶ Complex System (physics, ..., ethology, ecology, ...)
 - ▶ Artificial Life, Neural networks, ...,
 - ▶ Social Psychology, Sociology, Activity Theory, Economy, ...

Global View

Global picture (from AAMAS 2018, 2019 Topics of Interests)

Global View

- ▶ **Agent Theories and Models**: Belief-Desire-Intention theories and models, Cognitive models, Models of emotions, ...
- ▶ **Communication and Argumentation**: Commitments, Communication languages and protocols, Speech act theory, Argumentation-based dialogue and protocols, ...
- ▶ **Agent Cooperation**: Biologically-inspired approaches and methods, Collective intelligence, Distributed problem solving, Teamwork, team formation, teamwork analysis, Coalition formation, ...
- ▶ **Knowledge Representation and Reasoning**: Ontologies for agents, Reasoning in agent-based systems, Single and multi-agent planning and scheduling, Reasoning about action, plans and change, Reasoning about knowledge, beliefs, goals and norms, ...
- ▶ **Agent Societies and Societal issues**: Organizations and institutions, Social networks, Socio-technical systems, Normative systems, Values in MAS (privacy, safety, security, transparency, ...), Coordination and control models for multiagent systems, Trust and reputation, Policy, regulation and legislation, Self-organization
- ▶ ...

Action Domains

- ▶ Open and Decentralized AI Systems
 - ▶ Integration of intelligent applications, with humans, organizations and the physical world
 - ▶ Making them interoperate, interact, cooperate in a flexible and consistent manner with each other
- ▶ Problem solving
 - ▶ Modeling and solving problems by cooperation between local solvers
 - ▶ Installing top-down and/or bottom-up (emergent) solving process
- ▶ Multi-agent based simulation
 - ▶ Modeling and reproducing complex phenomena of interacting entities in the real world in order to understand or to explain their behavior

Technologies

- ▶ Agent Architectures and Theories
- ▶ Coalition formation mechanisms
- ▶ Multi-Agent Planning
- ▶ Agent Communication Languages, Interaction Protocols
- ▶ Auction mechanisms
- ▶ Negotiation strategies and mechanisms, Argumentation
- ▶ Electronic Institutions, Organisations, Norms
- ▶ Reputation, Trust
- ▶ Mono & multi-agent Learning
- ▶ Self-organisation, emergence, ...

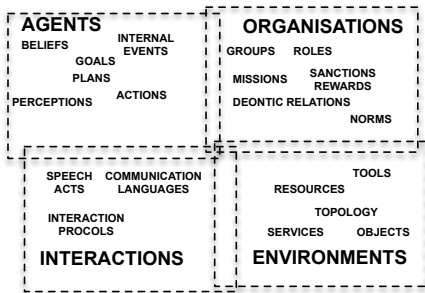
Current Landscape of Multi-Agent Programming

- ▶ Many Agent Oriented Software Engineering (AOSE) **methodology** (Prometheus, Gaia, Tropos, ...) exist!
- ↪ Use at least one of these methodologies for analysing and designing your MAS application
- ▶ Many agent languages have efficient and stable interpreters — used extensively in teaching
- ▶ All have some programming tools (IDE, tracing of agents' mental attitudes, tracing of messages exchanged, etc.)
- ▶ Some are integrating **social** aspects of MAS
- ▶ However, there are not yet proper tools to program multi-agent systems!
- ↪ some reasons and motivations follow!

Multi-agent programming: concepts

Multi-Agent Programming

VOWELS' perspective [Demazeau, 1995]:

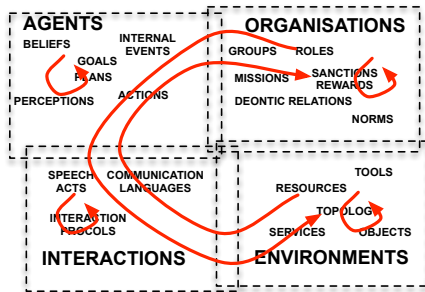


- ▶ **O**: abstractions for structuring and ruling the behaviours and interactions of agents in shared environment
- ▶ **A**: abstractions for the definition of the decision and reasoning architectures of autonomous entities
- ▶ **I**: abstractions for defining interactions among entities
- ▶ **E**: abstractions for defining and structuring resource/processing entities shared among the agents

→ A rich set of abstractions to address applications complexity!

Multi-agent programming: dynamics

Multi-Agent Programming



- ▶ Various life/control cycles among the concepts exist
 - ▶ Coordination of the system may be programmed using one or several families of concepts [Boissier, 2003]
- ↪ Interlacement of the various dynamics into bottom-up / top-down global cycles

↪ A rich palette of dynamics to address applications complexity!!

Multi-agent programming: approaches

Multi-Agent Programming

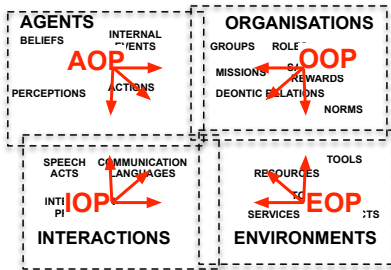
Agent Oriented
Programming

[Shoham, 1993]

Interaction
Oriented

Programming

[Huhns, 2001]



Organisation Oriented
Programming

[Pynadath et al., 1999]

Environment Oriented
Programming

[Ricci et al., 2010]

Multi-agent programming: platforms

Multi-Agent Programming

- ▶ Platforms not covering the whole set of abstractions:
 - ▶ JADE [Bellifemine et al., 1999, Bellifemine et al., 2000], Java-based Intelligent Agent Componentware [Lützenberger et al., 2013].
 - ▶ JACK [Winikoff, 2005, Howden et al., 2001] Jason [Bordini et al., 2007] 2APL [Dastani et al., 2003], GOAL [Hindriks, 2009] Jadex [Pokahr et al., 2005, Pokahr et al., 2014]
- ▶ Integrated programming approaches:
 - ▶ Volcano platform [Ricordel and Demazeau, 2002], MASK platform [Occello et al., 2004], MASQ [Stratulat et al., 2009], Situated E-Institutions [Campos et al., 2009], MANET [Tampitsikas et al., 2011], ANTE [Cardoso et al., 2016], Electronic Institutions - EI/EIDE [Noriega and de Jonge, 2016], InstAL [Padget et al., 2016], ROMAS/MAGENTIX2 [García et al., 2016], RTEC [Artikis et al., 2016], SARL [Rodriguez et al., 2014] BRAHMS[Sierhuis et al., 2003]
 - ↪ Socio-cognitive systems [Aldewereld et al., 2016]

However some families of concepts lose their control & visibility!

Agent concepts and approaches

Literature

Agent Basic Concepts

Books: [Bordini et al., 2005], [Bordini et al., 2009]

Proceedings: ProMAS, DALT, LADS, EMAS, AGERE, ...

Surveys: [Bordini et al., 2006], [Fisher et al., 2007] ...

Languages of historical importance: Agent0 [Shoham, 1993],
AgentSpeak(L) [Rao, 1996], MetateM [Fisher, 2005],
3APL [Hindriks et al., 1997],
Golog [Giacomo et al., 2000]

Other prominent languages:

Jason [Bordini et al., 2007], *Jadex* [Pokahr et al., 2005],
2APL [Dastani, 2008], GOAL [Hindriks, 2009],
JACK [Winikoff, 2005], JIAC, AgentFactory

But many other languages and platforms...

Some Languages and Platforms

Agent Basic Concepts

Jason (Hübner, Bordini, ...); 3APL and 2APL (Dastani, van Riemsdijk, Meyer, Hindriks, ...); Jadex (Braubach, Pokahr); MetateM (Fisher, Guidini, Hirsch, ...); ConGoLog (Lesperance, Levesque, ... / Boutilier – DTGolog); Teamcore/ MTDP (Milind Tambe, ...); IMPACT (Subrahmanian, Kraus, Dix, Eiter); CLAIM (Amal El Fallah-Seghrouchni, ...); GOAL (Hindriks); BRAHMS (Sierhuis, ...); SemantiCore (Blois, ...); STAPLE (Kumar, Cohen, Huber); Go! (Clark, McCabe); Bach (John Lloyd, ...); MINERVA (Leite, ...); SOCS (Torroni, Stathis, Toni, ...); FLUX (Thielscher); JIAC (Hirsch, ...); JADE (Agostino Poggi, ...); JACK (AOS); Agentis (Agentis Software); Jackdaw (Calico Jack); *simpAL*, *ALOO* (Ricci, ...);



Theories, Models, Architectures

Agent Basic Concepts

- ▶ Agents are used to solve problems (e.g. to find solutions, to take decisions, to act on the environment)
- ▶ The characteristics of the problem influence the way the agents are built
 - ↪ we then talk about **agent architectures**
- ▶ It may be the case that some architectures are designed using general principles
 - ↪ we then talk about **agent models**
- ▶ Some of these models have a theory associated with them that allows the verification of some properties
 - ↪ we then talk about **agent theories**

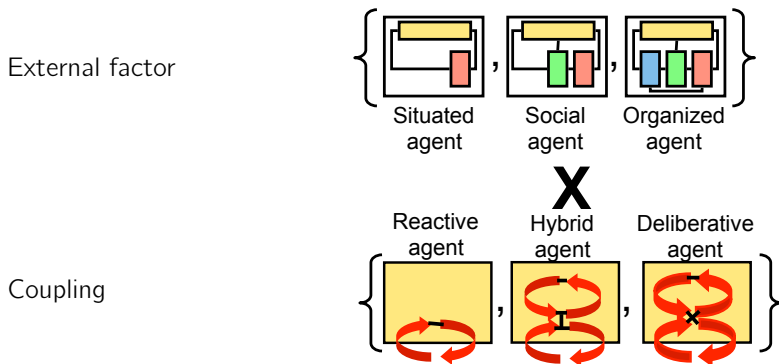
Several agent architectures, models and theories exist in the literature!!!

Analysis Grid

Panorama of Agent Models

Agent models depend on:

- ▶ the type of inputs that they reason from (external factor)
- ▶ the control cycle connecting inputs to actions (coupling)



External Factor

Panorama of Agent Models

- ▶ **Situated** Agents

- ▶ agents that reason about **themselves** and about their **environment**

- ▶ **Social** Agents

- ▶ agents that reason about themselves, about their environment and about the **interactions** with others

- ▶ **Organized** Agents

- ▶ agents that reason about themselves, about their environment and about the interactions with others and about the **organizations** (e.g. social structures, norms) enforcing these interactions

Coupling

Panorama of Agent Models

- ▶ **Reactive** Agent

- ▶ tight coupling between perception of the external factors with action

- ▶ **Deliberative** Agent

- ▶ loose coupling between perception and actions: agents deliberate on the actions to execute from their perception of the external factors and from their goals

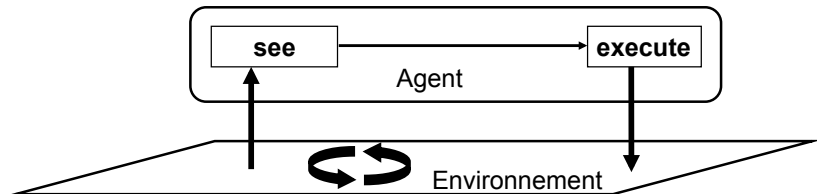
- ▶ **Hybrid** Agent

- ▶ agents that are mixing reactivity and deliberation

Reactive Agent Models

Panorama of Agent Models/ Coupling Dimension

- ▶ The process cycle of an agent is a closed loop between "execute" and "see" (Stimulus/Response)
- ▶ reaction to the evolution of the environment
- ▶ No explicit representation of the environment, of the other agents, of its skills,
- ▶ Decisions are done without reference to the past (no history), to the futur (no planning)



Reactive Agent models

Panorama of Agent Models/ Coupling Dimension

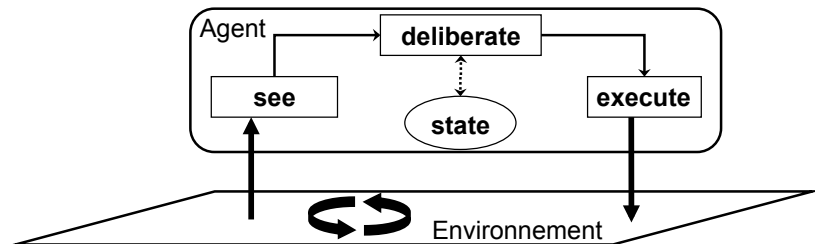
Reactive approach arises in opposition to the symbolic reasoning model (AI). Several approaches that are based on :

- ▶ behaviours
 - ▶ [Brooks, 1986], (Steels 89), (robotic)
 - ▶ (Drogoul 93) (ethology)
- ▶ interactions
 - ▶ (Demazeau 93) (image analysis, cartography, ...)
 - ▶ (Bura 91) (games)
- ▶ situations
 - ▶ (Agre 87) (games)
 - ▶ (Wavish 90) (design, manufacturing)

Deliberative Agent models

Panorama of Agent Models/ Coupling Dimension

- ▶ The process cycle of an agent introduces a "deliberate" function between "see" and "execute" in order to choose the "right" action
- ▶ Explicit Representation of the environment, of the other agents, of its skills, ...
- ▶ History management, ...



Deliberative Agent models

Panorama of Agent Models/ Coupling Dimension

- ▶ Goal-based Agents
 - ▶ Rich internal state
 - ▶ Can anticipate the effects of their actions (e.g. Planning)
 - ▶ Take those actions expected to lead toward achievement of goals
 - ▶ Capable of reasoning and deducing properties of the world (Knowledge representation)
- ▶ Utility-based Agent
 - ▶ Decision Theory + Probabilities
 - ▶ Use of utility function that maps state (or state sequences) into real numbers
 - ▶ Permits more fine-grained reasoning about what can be achieved, what are the trade-offs, conflicting goals, etc

Hybrid Agent Models

Panorama of Agent Models/ Coupling Dimension

Hybrid Agent's Model: Reactive and Deliberative Agent

- ▶ Reactive agents are too simple - they work well in some scenarios, but they fail to solve complex problems
- ▶ Deliberative agents are too complex - they need too much time to deliberate, they fail in very dynamic environments
- ▶ The reactive and deliberative behaviors are organized in layers
- ▶ Examples: Touring Machines [Ferguson, 1995], InterRaP[Müller and Pischel, 1994],

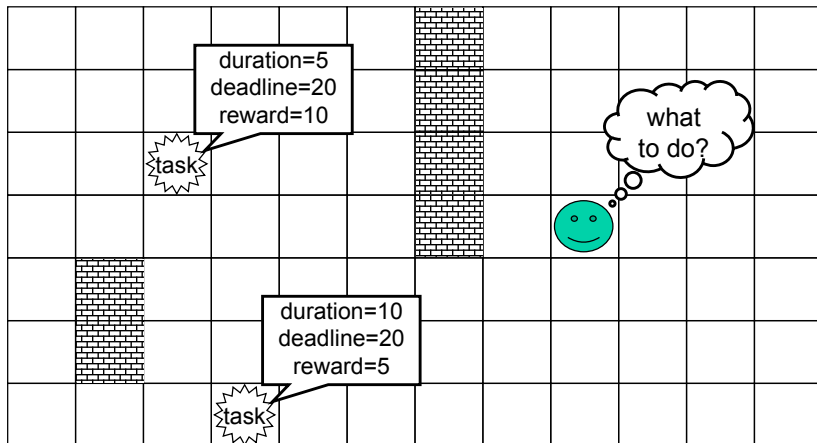
Situated Agent

Panorama of Agent Models

- ▶ **Reactive** agents: the subsumption architecture [Brooks, 1986]
- ▶ **Deliberative** agents: the BDI model and the PRS architecture
- ▶ **Hybrid** agents: Touring Machines [Ferguson, 1995]
- ▶ Reason about themselves and about their environment
- ▶ We need to model the environment (subject of the **Agent working environment** course)
- ▶ Our case study:
 - ▶ the agents move on a 2D grid
 - ▶ there are obstacles blocking their movements
 - ▶ an agent should find a path to a task, to execute it, and then to move on to another task
- ▶ Note: movement on a grid stands for real movement (e.g., robots) or virtual movement (e.g., searching on Internet)

Case study

Panorama of Agent Models/ Situated Agent



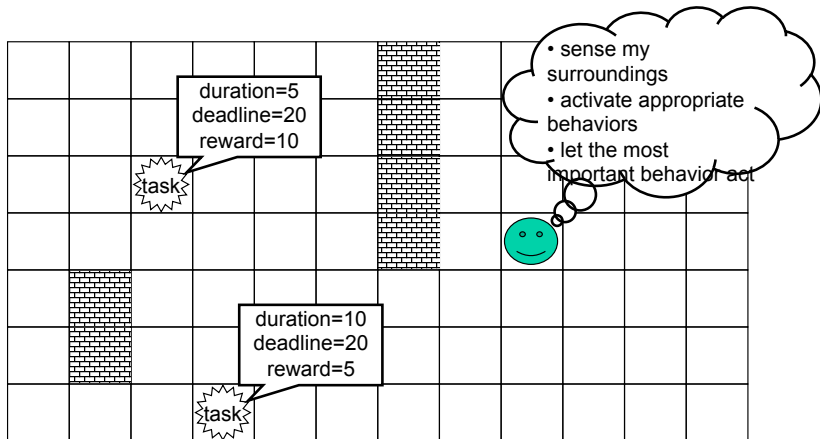
The Subsumption Architecture

Panorama of Agent Models/ Situated Agent/ **Reactive agents**

- ▶ Agent's decision making is realized through a set of tasks accomplishing behaviors.
- ▶ A behavior continually takes perceptual inputs and maps them to an action to perform (finite state machines, no symbolic reasoning, no symbolic representation)
- ▶ Many behaviors can fire simultaneously. In order to choose between them, use of a subsumption hierarchy, with the behaviors arranged into layers.
A higher layer has priority on lower layers (inhibition)

The Subsumption Architecture

Panorama of Agent Models/ Situated Agent/ Reactive agents



The Subsumption Architecture

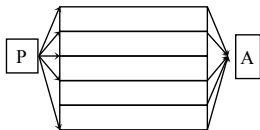
Panorama of Agent Models/ Situated Agent/ Reactive agents

- ▶ Does it work? The agents are very simple, there is no symbolic reasoning or representation of their environment...
- ▶ It works if there are many agents: “the intelligence is in the system, not in the entities composing it”.
- ▶ (Steels 89) used this architecture in a scenario very similar with our case study:
 - ▶ robots have to collect samples of precious rock (unknown location) and bring them back to a mothership spacecraft.
 - ▶ cooperation without direct communication : through the environment.
 - ▶ gradient field with a signal generated by the mothership
 - ▶ radioactive crumbs are picked up, dropped and detected by robots.

Agent Architectures

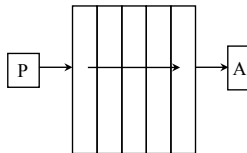
Panorama of Agent Models

► Modules Organisation:

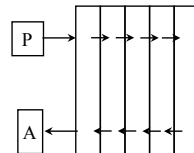


P : perception, A : action

a) horizontal architecture



b) modular vertical architecture
one path

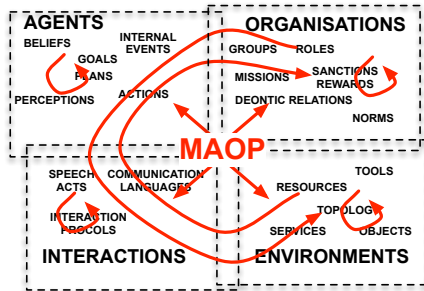


c) layered vertical architecture
two paths

- Control flow: one / several
- Data flow: broadcast, translation
- Control structure: inhibition, hierarchy, ...

Multi-Agent Oriented Programming

Multi-Agent Oriented Programming



Multi-Agent Oriented Programming (MAOP) aims at programming systems:

- ▶ as **organisation** of autonomous **agents** in **interaction** with each other within a shared **environment**,
- ▶ by keeping alive from design to execution, concepts pertaining to each of the A/E/I/O families as well as their control/life cycles.

~> Going beyond each of the A/E/I/O oriented programming approaches

Key features

Multi-Agent Oriented Programming

▶ **Abstraction**

- ▶ keeping the concepts alive from design to execution, e.g. no agents sharing and calling OO objects
- ▶ effective programming models for controllable and observable computational entities

▶ **Modularity**

- ▶ away from the monolithic and centralised view

▶ **Orthogonality**

- ▶ wrt models, architectures, platforms
- ▶ support for heterogeneous systems

▶ **Dynamic extensibility**

- ▶ dynamic construction, replacement, extension of the entities participating to the system
- ▶ support for open systems

▶ **Reusability**

- ▶ reuse of the entities participating to the system for different kinds of applications

JADE

Content

- Briefly about components
- JADE overview
- Setting up the Development Environment
 - Installing JADE
 - Installing Eclipse
 - Setting Eclipse to work with JADE
- Book Trading example

Components, Java



<http://java.sun.com/>

- Java technology is a portfolio of products that are based on the power of networks and the idea that the same software should run on many different kinds of systems and devices.
- The Java programming language lets you write powerful, enterprise-worthy programs that run in the browser, from the desktop, on a server, or on a consumer device.

Components, JADE



<http://jade.tilab.com/>

- JADE (Java Agent DEvelopment Framework) is a software framework fully implemented in Java language.
- It simplifies the implementation of multi-agent systems through a middle-ware that claims to comply with the FIPA specifications and through a set of tools that supports the debugging and deployment phase.

JADE overview

JADE is a middleware that facilitates the development of multi-agent systems. It includes

- A **runtime environment** where JADE agents can “live” and that must be active on a given host before one or more agents can be executed on that host.
- A **library** of classes that programmers have to/can use (directly or by specializing them) to develop their agents.
- A suite of **graphical tools** that allows administrating and monitoring the activity of running agents.

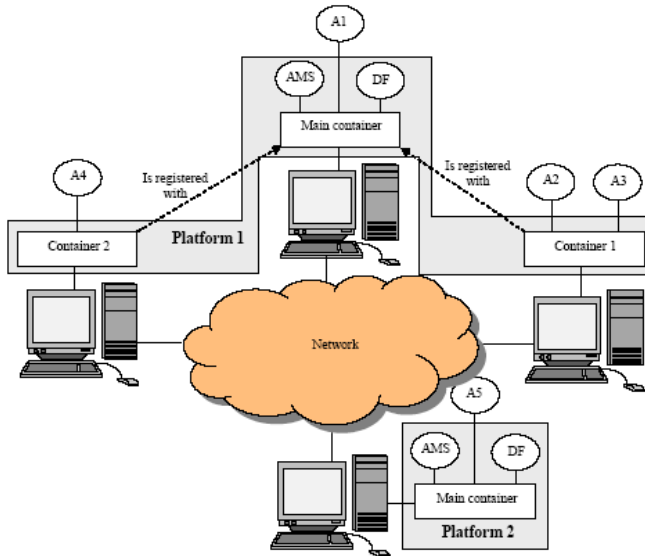
Containers and Platforms

- Each running instance of the JADE runtime environment is called a ***Container*** as it can contain several agents.
- The set of active containers is called a ***Platform***.
- A single special ***Main container*** must always be active in a platform and all other containers register with it as soon as they start.

- The **AMS** (Agent Management System) that provides
 - the naming service (i.e. ensures that each agent in the platform has a unique name)
 - represents the authority in the platform (for instance it is possible to create/kill agents on remote containers by requesting that to the AMS).

- The **DF** (Directory Facilitator) that provides a Yellow Pages service by means of which an agent can find other agents providing the services he requires in order to achieve his goals.

Sample architecture



Getting JADE

All the software is distributed under the LGPL license limitations and it can be downloaded from the JADE web site <http://jade.tilab.com/>. Five compressed files are available:

- The source code of JADE
- The source code of the examples
- The documentation, including the javadoc of the JADE API and this programmer's guide
- The binary of JADE, i.e. the jar files with all the Java classes
- A full distribution with all the previous files



CONCLUSION

Hybrid agents

Panorama of Agent Models/ Situated Agent/ Hybrid agents

- ▶ Reactive agents are too simple - they work well in some scenarios, but they fail to solve complex problems
- ▶ Deliberative agents are too complex - they need too much time to deliberate, they fail in very dynamic environments
- ▶ Solution: hybrid agents that are both reactive and deliberative, depending on the situation.
- ▶ The reactive and deliberative behaviors are organized in layers $\leadsto$ layered architectures.

Touring Machines

Panorama of Agent Models/ Situated Agent/ Hybrid agents

- ▶ Constrained navigation in dynamic environments
- ▶ Consists of three activity producing layers : each layer produces suggestions for the actions to perform.
 - ▶ Reactive layer: reactive behaviour
 - ▶ Planning Layer: proactive behaviour
 - ▶ Modeling Layer: world updates, beliefs; it predicts conflicts between agents and it changes the plans/goals
- ▶ Control-subsystem: chooses the active layer: certain observations should never reach certain layers.

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